FEATURE ENGINEERING & PREPROCESSING STATUS REPORT

Disaster Response Classifier · HumAID, Kaggle & CrisisLexT26

Prepared 17 October 2026 | Branch: feature/preprocessing

1. Executive summary

Preprocessing implementation and artifacts have been prepared and handed off. Reported verification: 8 passing unit tests and a clean git diff check. Prepared components include HumAID TF-IDF, BiLSTM sequences, BERTweet inputs, numerical features, a separate Kaggle binary preprocessing path, and CrisisLexT26 transformations using frozen HumAID representations. Model training and performance evaluation have not been performed as part of this task.

2. Deliverables and dimensions

Component	Evidence / dimensions
HumAID splits	Train 53,531; validation 7,793; test 15,160; total 76,484.
TF-IDF	50,000 features for HumAID train/validation/test.
BiLSTM	Vocabulary 29,351; minimum frequency 2; <PAD>/<UNK>; length 64.
BERTweet	vinai/bertweet-base; max length 256; attention masks; tokenizer saved.
Numerical features	8 features; scaler fitted on training data only.
Kaggle	6,090/761/762 rows; 15,461 TF-IDF features.
CrisisLexT26	27,933 rows; frozen HumAID representations; 28.39% BiLSTM OOV.
Verification	8 unit tests passed; git diff --check passed.
Git/Drive	Branch feature/preprocessing pushed; user reports Drive upload of both ZIPs and README.
Model training	Not performed as part of this preprocessing assignment.

3. Checklist status legend

DONE = supported by recorded implementation/audit. PARTIAL / VERIFY = implementation exists but a requested check is not evidenced. NOT VERIFIED = no reliable evidence available in handoff notes. These statuses identify remaining sign-off checks; they do not assert that the underlying item is necessarily broken.

A. Universal preprocessing

Checklist item	Status	Evidence / note
Separate cleaning/tokenization by model type	**DONE**	Separate TF-IDF, BiLSTM sequence and BERTweet paths prepared.
HTML entity unescaping explicitly verified on fresh samples	**PARTIAL / VERIFY**	Cleaning is implemented, but fresh-sample verification across every dataset/model path is not evidenced.
URL and mention handling	**DONE / MODEL-DEPENDENT**	Placeholder/normalization handling is used in conventional text cleaning; BERTweet preserves more original tweet structure.
Strip retweet prefixes	**DONE / MODEL-DEPENDENT**	Retweet-prefix handling is documented in the cleaning path.
Hashtag, emoji, elongated-text strategy	**PARTIAL**	Hashtag words are retained in TF-IDF; BERTweet preserves casing, emojis, punctuation and informal spellings. Cross-dataset rule parity needs verification.
Casing and punctuation per model	**DONE**	TF-IDF is normalized/lowercased; BERTweet preserves original casing and punctuation.
Deterministic preprocessing	**DONE / TESTED**	8 unit tests passed.
Sequence lengths selected from EDA percentiles	**PARTIAL / VERIFY**	BiLSTM length 64 and BERTweet max length 256 implemented; percentile-based justification is not confirmed in available evidence.
No inference-time feature leakage	**PARTIAL / VERIFY**	Train-only fitting is implemented for key representations; a full feature-by-feature leakage review is not evidenced.
Shared importable preprocessing functions	**PARTIAL / VERIFY**	Preprocessing code is in the repository; verify all downstream scripts import the shared functions.

B. HumAID-specific

Checklist item	Status	Evidence / note
TF-IDF fit on train only; transform validation/test	**DONE**	Train 53,531×50,000; validation 7,793×50,000; test 15,160×50,000.
Train-only BiLSTM vocabulary, cutoff and <UNK>	**DONE**	Vocabulary 29,351; min frequency 2; <PAD>/<UNK>; length 64.
BERTweet tokenizer and max length	**DONE / PARTIAL**	vinai/bertweet-base tokenizer saved; max length 256 and attention masks. EDA percentile justification needs verification.
Reusable HumAID label2id/id2label mapping for 11 classes	**PARTIAL / VERIFY**	Supplied data contains 10 actual classes. Confirm reusable mapping and expected schema with project owner; do not invent an 11th class.
OOV rate logged	**PARTIAL**	CrisisLex BiLSTM OOV is 28.39%; HumAID-specific OOV report not confirmed.
Exclude event metadata from model inputs	**PARTIAL / VERIFY**	Confirm final model input columns exclude event metadata.
Save vectorizer/tokenizer/vocabulary artifacts	**DONE**	Saved under models/preprocessing/ and included in Drive handoff ZIP.
Fresh-sample HTML entity verification	**PARTIAL / VERIFY**	Implementation exists; specific verification record is not confirmed.

C. Kaggle binary pre-filter

Checklist item	Status	Evidence / note
Separate Kaggle feature pipeline	**DONE / VERIFY FIT SOURCE**	6,090/761/762 train/validation/test rows; 15,461 TF-IDF features. Confirm fitting source in code.
Keyword feature decision documented	**PARTIAL / VERIFY**	Exact final decision/rationale not confirmed.
Location feature decision documented	**PARTIAL / VERIFY**	Exact final decision/rationale not confirmed.
Binary target encoding separate from HumAID labels	**PARTIAL / VERIFY**	Separate binary task prepared; verify exact target encoding in code/artifacts.
Same cleaning rules applied consistently	**PARTIAL / VERIFY**	Separate Kaggle path exists; full rule-by-rule cross-dataset equivalence not evidenced.
Distinct Kaggle artifacts	**DONE**	Stored separately under data/processed/kaggle/.
Only labeled train.csv used for fitting	**PARTIAL / VERIFY**	Outputs exist; source-by-source fit provenance needs code check.

D. CrisisLexT26-specific

Checklist item	Status	Evidence / note
Do not fit a new vectorizer/vocabulary on CrisisLex	**DONE**	Transformed using frozen HumAID representations.
Reuse exact HumAID cleaning function	**PARTIAL / VERIFY**	Frozen representations reused; literal shared-function reuse needs confirmation.
ID-based text rehydration where applicable	**NOT VERIFIED**	No confirming evidence in current handoff notes.
Map labels and flag uncertain rows	**PARTIAL**	10,959 good mappings; 13,555 rough; 3,419 unmapped. Review rough/unmapped rows.
Measure OOV against frozen vocabulary	**DONE**	BiLSTM OOV reported as 28.39%.
Preserve per-event identity	**NOT VERIFIED**	Per-event breakdown preservation not confirmed.

E. Cross-dataset consistency

Checklist item	Status	Evidence / note
CrisisLex dimensions match HumAID	**DONE**	CrisisLex TF-IDF 27,933×50,000; BiLSTM 27,933×64; BERTweet 27,933×256.
Same padding/truncation rule	**DONE / VERIFY**	Lengths match (64 and 256); verify loaders/masks in downstream code.
Five-row side-by-side cleaning spot check	**NOT VERIFIED**	No recorded result found in handoff notes.
CrisisLex uses exact HumAID integer label IDs	**PARTIAL / VERIFY**	Mapping audit exists; exact integer-ID equivalence needs confirmation.
Prevent Kaggle vectorizer/HumAID path mix-up	**DONE / VERIFY**	Artifacts are separated; verify explicit filenames in loaders.
Versioned artifacts loaded by explicit filename	**PARTIAL**	Artifacts separated by directory; explicit version metadata/loading needs review.

4. Data integrity and known limitations

• HumAID supplied files contain 10 actual classes although checklist language mentions 11. Confirm intended schema; do not fabricate a class.

• Four exact duplicate tweet texts were found across HumAID train and validation; none across train/test or validation/test. Official split files were not modified.

• Near-duplicate/retweet-chain leakage and event-level leakage were not established; the audit evidence did not provide event IDs.

• CrisisLexT26 label mapping: 10,959 good, 13,555 rough, 3,419 unmapped. Review before using uncertain labels for supervised evaluation.

• A suspicious malformed/encoding-uncertain Kaggle text sample was left unchanged and documented.

• Processed artifacts are ignored by Git and were shared separately through Drive ZIPs; a fresh Git clone will not automatically contain them.

5. GitHub and Drive handoff

• GitHub branch: feature/preprocessing

• Branch: https://github.com/Abhishek-Choudhary15/Disaster_Response_Classifier/tree/feature/preprocessing

• PR page: https://github.com/Abhishek-Choudhary15/Disaster_Response_Classifier/pull/new/feature/preprocessing

• Drive upload reported by user: processed_data.zip, preprocessing_artifacts.zip, README.txt. Share the actual Drive folder link separately.

• Processed artifacts are ignored by Git and shared separately. Do not stage or modify src/models/.

6. Remaining actions before final sign-off

1. Verify HumAID label2id/id2label mapping and row-order alignment between labels and all feature representations.

2. Resolve all PARTIAL / VERIFY and NOT VERIFIED items above before claiming 100% checklist completion.

3. Review the four train/validation duplicate pairs and document the chosen treatment without silently changing official splits.

4. Review rough/unmapped CrisisLex labels and document inclusion/exclusion criteria.

5. Ensure teammates download both ZIPs and extract into expected project-relative directories.

6. Proceed to model training/evaluation only after artifact and label alignment checks.

7. Scope boundary

This report covers feature engineering and preprocessing only. It does not claim that Perceptron, MLP, BiLSTM, BERTweet fine-tuning, model comparison, final test evaluation, external robustness evaluation, or deployment has been completed.